1. Project Planning & Management

- **Project Proposal – Overview of the project, objectives, and scope.**

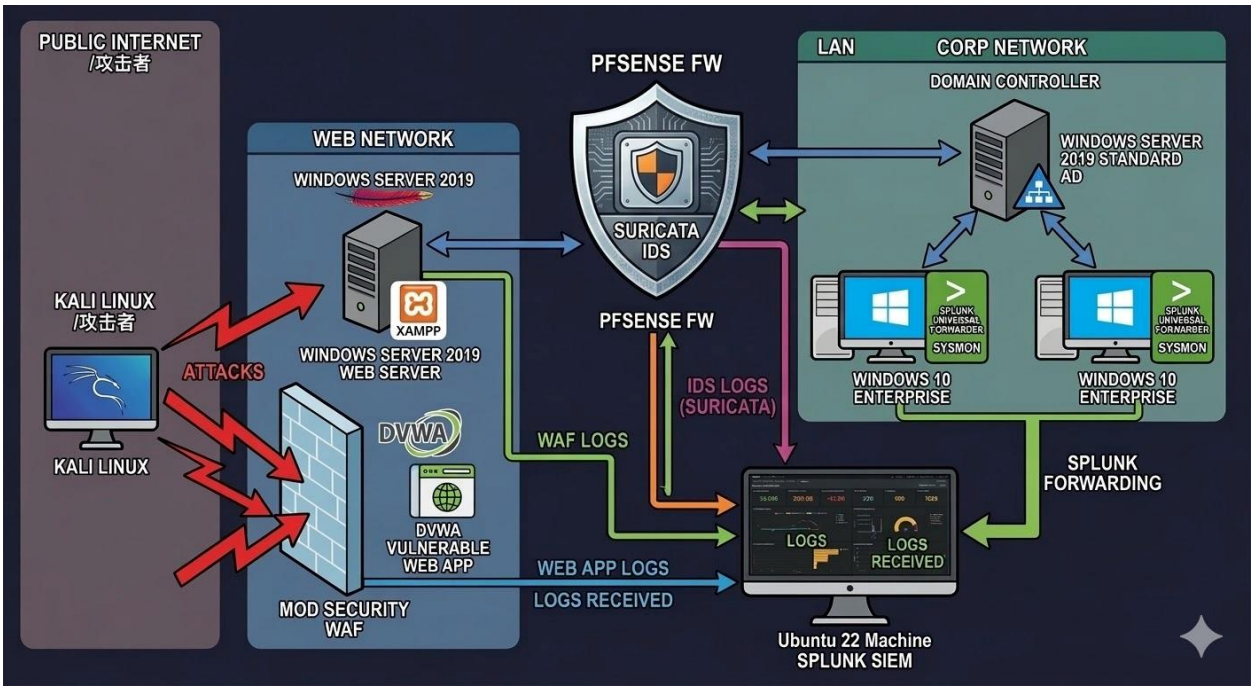
Project 1: Building and Operating a Mini Security Operations Center (SOC)

- **Project Title:** Security Operations Center (SOC) Implementation and Threat Detection.

Objectives:
1 - Simulate a full SOC workflow including log ingestion, threat detection, triaging, and reporting.
2 - To build a functional SOC environment using Splunk for monitoring, a pfSense firewall for network security, and various detection tools like Suricata and ModSecurity.

- **Project Plan – Timeline (Gantt chart), milestones, deliverables, and resource allocation.**

Project Plan:



Timeline (Gantt Chart View):

This timeline follows the 4-week sprint structure suggested in project booklet for building the SOC and performing incident response.

Week 1: SOC Setup & Log Ingestion:

Environment Setup (March)

Tasks:

- Set up a centralized logging server using Splunk.
  - Install Splunk Enterprise on Ubuntu.

- Configure log sources: Windows, firewall, and IDS.
  - Deploy pfSense Firewall and configure LAN/WAN interfaces.
  - Set up Windows Domain Controller.

Deliverables:

- Operational log ingestion pipeline.
- Documented SOC architecture.

Week 2: SIEM Configuration & Use Case Development:

Data Ingestion & Security Layers (April)

- Install Universal Forwarders and Sysmon on endpoints.
- Integrate Suricata (IDS/IPS) and ModSecurity (WAF).
- Verify log flow from all sources into Splunk.

Tasks:

- Deploy SIEM platform Splunk.
- Configure 3–5 real-world use cases (e.g., brute force, malware infection).

Deliverables:

- Use case documentation with correlation rules.
- Screenshots of triggered alerts.

Week 3: Alert Triage and Incident Management:

Detection & Simulation (Early May)

- Configure SPL rules and alerts in Splunk.
- Simulate attacks (SQLi for WAF, testmyids.com for Suricata).
- Perform Active Directory attack simulations.

Tasks:

- Triage and analyze triggered alerts.
- Apply MITRE ATT&CK mapping to events.

Deliverables:

- Triage sheets and IOC identification.
- Evidence of containment or analysis recommendations.

Week 4: Reporting & Final Presentation:

Documentation & Hardening (Mid May)

- Draft the Incident Response Playbook.
- Finalize the Technical Report and Executive Summary.
- Record the project demonstration video.

Tasks:

- Prepare KPI metrics.
- Perform root cause analysis for a selected incident.

Deliverables:

- Final Report: SOC architecture, SIEM rules, KPIs, analysis.
- Presentation: SOC design, use cases, response strategy.

Project Milestones & Deliverables:

This table outlines the major phases of SOC project.

Milestone	Deliverable	Due Date
M1: Planning & Scope	Project Proposal & Risk Assessment	Jan 27, 2026
M2: Research & Req.	Literature Review & Hardware/Software Specs	Mar 1, 2026
M3: System Design	Network Topology & SIEM Architecture Design	Apr 3, 2026
M4: Implementation	Functional SOC (Splunk, pfSense, WAF, IDS)	May 17, 2026
M5: Final Submission	Final Report, Test Cases, & Presentation	May 24, 2026

Deliverables:

1. A fully configured Splunk Enterprise instance.
2. Active Directory and Windows log forwarding.
3. Web Application Firewall (WAF) integration.
4. Intrusion Detection/Prevention System (IDS/IPS) setup.

Resource Allocation:

distribution of responsibilities:

Role	Responsibility	Primary Tools
SOC Architect	Infrastructure setup, pfSense configuration, and networking.	VMware, pfSense
SIEM Administrator	Splunk installation, Index management, and Syslog-ng setup.	Splunk, Linux (Ubuntu)
Security Engineer	IDS/IPS (Suricata) and WAF (ModSecurity) implementation.	Suricata, Apache/XAMPP
Endpoint Specialist	Sysmon deployment, GPO configuration, and Windows logging.	Windows Server, Sysmon
Threat Hunter	Writing SPL queries, creating alerts, and attack simulation.	Splunk (SPL), Kali Linux
Technical Writer	Documentation, Gantt chart, and Final Presentation.	GitHub, LaTeX/Office

Task Assignment & Roles – Defined responsibilities for team members.

Task	Assignment	responsibilities
AD environment and, Sysmon deployment, GPO configuration, Windows logging	Abdalaziz Mohammad Mohammad	**All responsible for his:** 1- Infrastructure setup, configuration, and networking. 2- Index management, and Syslog-ng setup. 3- Writing SPL queries, creating alerts, and attack simulation. 4 - collect his logs and send it to Splunk. 5 - create use cases and correlate it. 6 - make Documentation, Gantt chart, and Final Presentation (for his specific task according to the previous schedule).
Splunk installation	Ahmed Hamdy	
PFSENSE FW	Marko Maged	
SURICATA IDS/IPS	Ahmed Atef	
XAMP WEB SERVER AND DVWA WEB APPLICATION	Bassam Alsayed	
WAF (ModSecurity)	Ahmed Tarek	

Tasks Sequence:

1- install splunk.
2- install forwarder.
3- configure Domain Controller Environment.
4- install Xampp Web app server.
5- install Pfsense firewall.
6- install Suricata IPS/IDS.
6- configure DVWA vulnerable Web app on xampp.
7- configure securityMod WAF To detect web app attacks.
8- write rule in splunk to generate alert.
9- detect Active Directory Attacks.